

JSS1 DIGITAL TECHNOLOGIES LESSON NOTES

SECOND TERM

WEEK 1: MS POWERPOINT (INTRODUCTION)

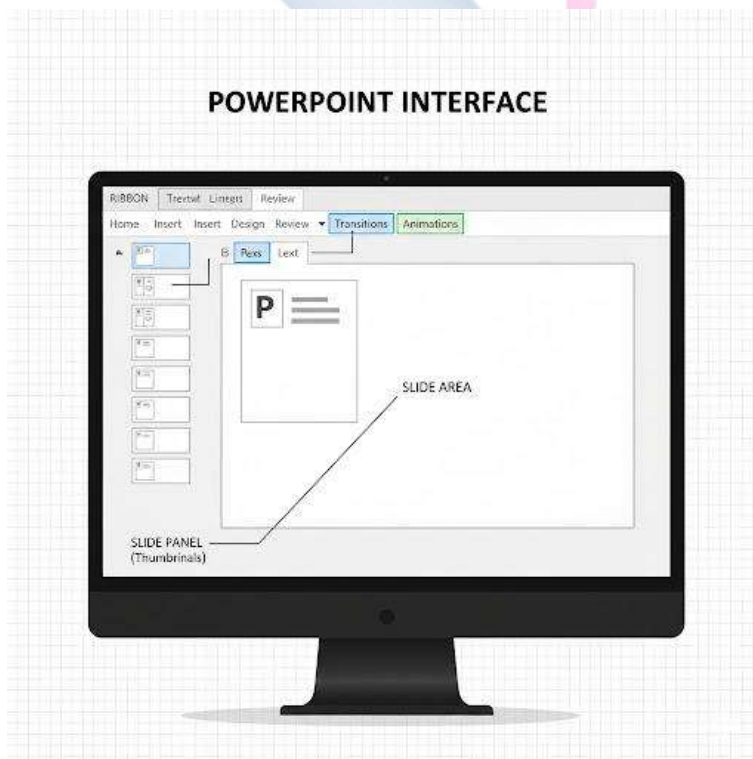
SUBTOPICS:

- MEANING AND USES OF PRESENTATION SOFTWARE
- CREATING AND SAVING SLIDES
- ADDING TEXT, IMAGES, TRANSITIONS

WHAT IS MICROSOFT POWERPOINT?

PowerPoint is a presentation software used to create slideshows that combine text, images, and effects.

Other Presentation Software: Google Slides, Apple Keynote, Canva



USES OF POWERPOINT:

- Teachers present lessons
- Students present projects
- Business meetings
- Photo slideshows
- Church presentations

PARTS OF POWERPOINT WINDOW:

1. **Title Bar:** Shows presentation name
2. **Ribbon:** Contains tools (Home, Insert, Design, Transitions, Animations)
3. **Slide Panel (Left):** Shows all slides
4. **Slide Area (Center):** Where you design
5. **Notes Section (Bottom):** Speaker notes

OPENING POWERPOINT:

- Start menu → Microsoft PowerPoint
- OR Desktop shortcut

CREATING NEW PRESENTATION:

1. Open PowerPoint (blank presentation appears)
2. OR File → New → Blank Presentation

WORKING WITH SLIDES:

Add New Slide: Home tab → New Slide (or Ctrl + M) **Delete Slide:** Click slide → Press Delete
Move Slides: Drag slide up/down in left panel **Copy Slide:** Right-click → Duplicate Slide

ADDING TEXT:

1. Click "Click to add title" or "Click to add text"
2. Type your content
3. Format using Home tab (Font, Size, Bold, Color)

ADDING IMAGES:

1. Insert tab → Pictures
2. Choose "This Device" or "Online Pictures"
3. Select image → Insert
4. Resize by dragging corners

ADDING SHAPES:

1. Insert tab → Shapes
2. Choose shape
3. Click and drag to draw

DESIGN AND THEMES:

1. Design tab → Choose theme
2. Applies colors, fonts, and backgrounds to all slides

TRANSITIONS:

What are Transitions? Effects when moving from one slide to another.

Adding Transitions:

1. Select slide

2. Transitions tab → Choose effect (Fade, Push, Wipe)
3. Set duration (speed)
4. "Apply To All" for all slides

ANIMATIONS:

What are Animations? Effects for objects on slides (text, images).

Adding Animations:

1. Select object
2. Animations tab → Choose animation (Appear, Fly In, Fade)

SAVING PRESENTATION:

First Time:

1. File → Save As
2. Choose location
3. Type file name → Save

After Editing: Ctrl + S

RUNNING SLIDESHOW:

- Slide Show tab → From Beginning
- OR Press F5
- Click or Spacebar for next slide
- Esc to exit

TIPS FOR GOOD PRESENTATIONS:

1. Keep text simple (short sentences)
2. Use large fonts (at least size 24)

3. Choose clear colors
4. Don't overcrowd slides
5. Use relevant images
6. Practice before presenting

EVALUATION:

1. What is PowerPoint used for?
2. How do you add a new slide?
3. What is the difference between transitions and animations?
4. Name three parts of the PowerPoint window.
5. How do you save a presentation?

ASSIGNMENT:

1. Explain what presentation software is with examples.
2. Describe how to add text and images in PowerPoint.
3. What are transitions? How do you add them?
4. List five tips for creating good presentations.

WEEK 2: 3D MODELLING WITH PAINT 3D

SUBTOPICS:

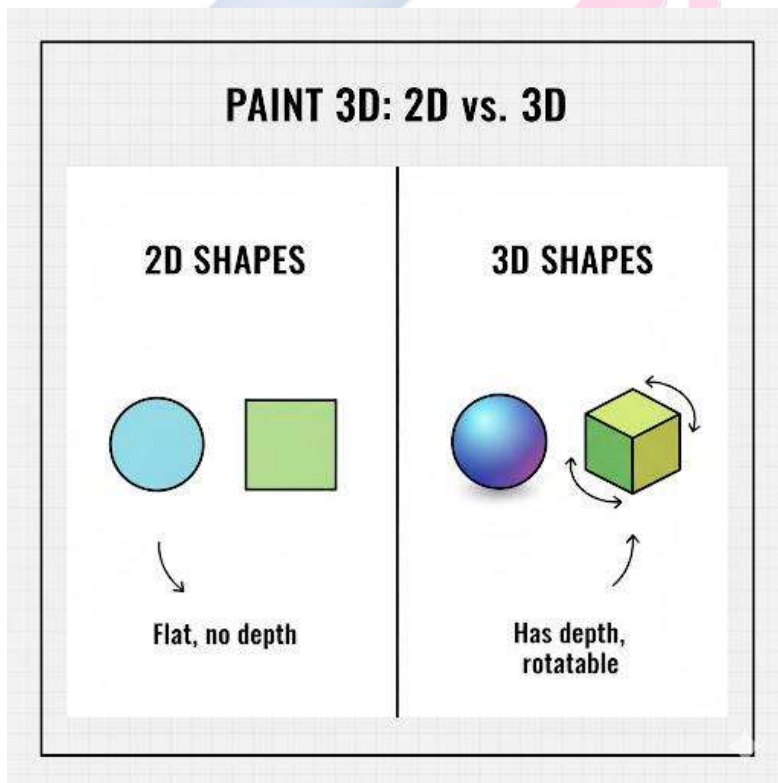
- INTRODUCTION TO PAINT 3D
- DRAWING BASIC SHAPES AND MODELS
- SAVING AND EXPORTING 3D PROJECTS

WHAT IS PAINT 3D?

Paint 3D is a free Microsoft app for creating 2D images and 3D objects.

Difference from Regular Paint:

- Regular Paint: Only flat (2D) drawings
- Paint 3D: Can create 3D objects you can rotate



OPENING PAINT 3D: Start menu → Type "Paint 3D" → Open

PARTS OF PAINT 3D:

1. **Canvas:** Center workspace
2. **Tool Menu (Left):** Brushes, 2D shapes, 3D shapes, Stickers, Text, Effects
3. **Color Picker (Right):** Choose colors and materials

2D VS 3D:

- **2D:** Flat (like paper)
- **3D:** Has depth (like real objects you can rotate)

USING BRUSHES:

1. Click Brushes icon
2. Choose brush type (Pencil, Marker, Watercolor)
3. Select color
4. Draw on canvas

2D SHAPES:

1. Click "2D shapes"
2. Choose shape (Circle, Square, Star)
3. Click and drag on canvas
4. Customize fill color, line color, thickness

3D SHAPES:

1. Click "3D shapes"
2. Choose shape (Cube, Sphere, Cylinder, Cone, Pyramid)
3. Shape appears on canvas

Working with 3D Objects:

- **Move:** Click and drag
- **Rotate:** Use rotation circles
- **Resize:** Drag corner handles
- **Color:** Click color on right panel
- **Material:** Choose Matte, Gloss, or Metal

CREATING SIMPLE MODELS:

Example 1: Snowman

1. Add 3 spheres (large, medium, small)
2. Stack vertically
3. Color white
4. Add small spheres for eyes/buttons
5. Add cone for nose (orange)

Example 2: Simple House

1. Add cube (house body)
2. Add pyramid on top (roof)
3. Add small rectangles (windows/door)
4. Color each part

STICKERS:

- Ready-made images you can add
- Click Stickers → Choose sticker → Place on canvas

TEXT:

- **2D Text:** Flat text
- **3D Text:** Text you can rotate

Steps:

1. Click Text icon
2. Choose 2D or 3D
3. Type text
4. Format (font, size, color)

SAVING YOUR PROJECT:

1. Menu (top left) → Save or Save As
2. Choose location
3. Name file → Save
4. Saves as .3mf file

EXPORTING:

As Image:

1. Menu → Save as → Image
2. Choose PNG or JPEG
3. Save

As Video:

1. Menu → Save as → Video
2. Creates rotating video of your 3D model

PROJECT IDEAS:

- Solar system (different sized spheres)
- Robot (cubes and cylinders)
- Birthday cake (stacked cylinders)
- Simple car or airplane

EVALUATION:

1. What is Paint 3D used for?
2. Name five 3D shapes available.
3. How do you rotate a 3D object?
4. What is the difference between 2D and 3D?
5. How do you save your project?

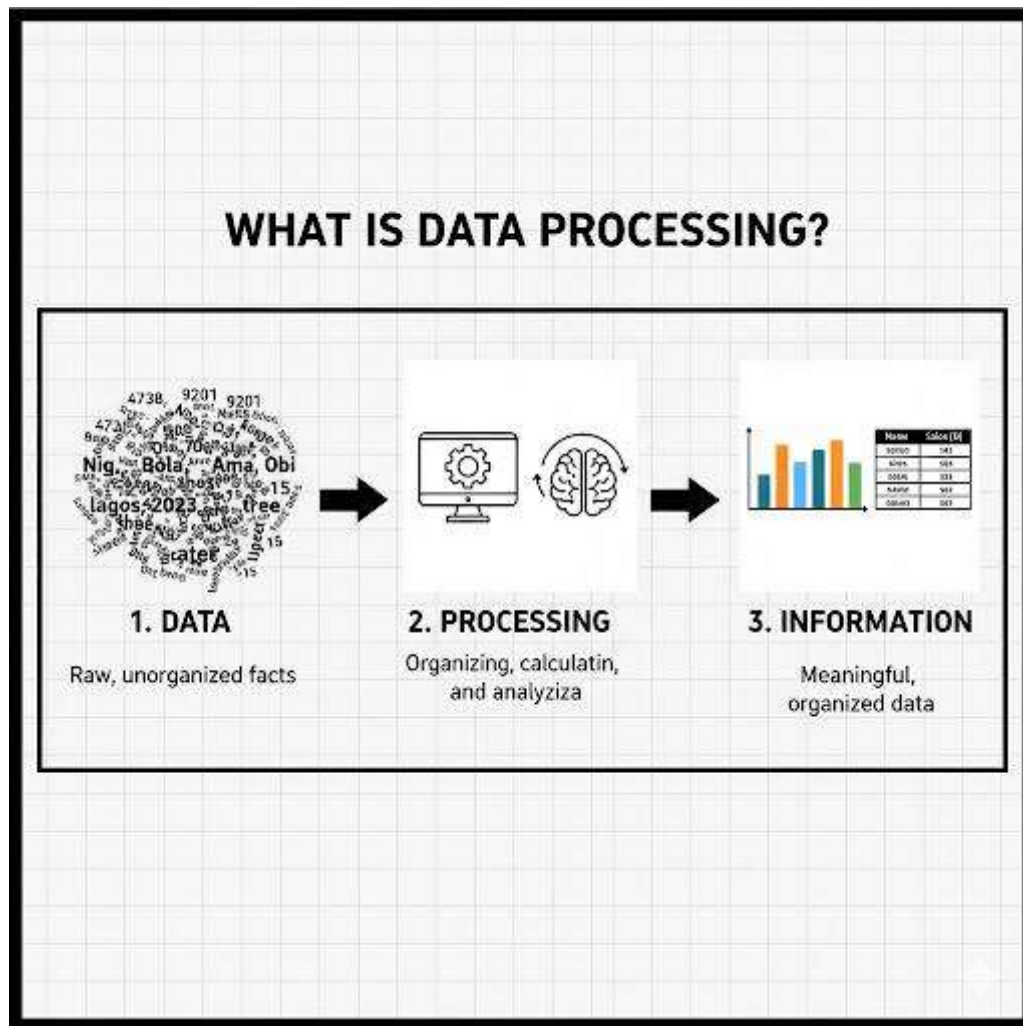
ASSIGNMENT:

1. Explain Paint 3D and how it differs from regular Paint.
2. List the materials available for 3D objects.
3. Describe how to create a simple snowman.
4. How do you export a project as an image?

WEEK 3: DATA PROCESSING

SUBTOPICS:

- MEANING AND STAGES OF DATA PROCESSING
- MANUAL VS ELECTRONIC DATA PROCESSING
- IMPORTANCE IN REAL-LIFE APPLICATIONS



MEANING OF DATA PROCESSING

Data: Raw facts without meaning (names, numbers, dates) **Information:** Processed data with meaning

Example:

- Data: John, 78
- Information: John scored 78 marks in Mathematics

Data Processing: Converting raw data into useful information.

STAGES OF DATA PROCESSING:**1. DATA COLLECTION (Input)**

- Gathering raw data
- Methods: Forms, surveys, keyboard, scanners
- Example: School collects student names and ages

2. VALIDATION

- Checking if data is correct
- Ensures accuracy
- Example: Age can't be 200 or letters

3. PROCESSING

- Performing operations on data
- Operations: Sorting, calculating, grouping, searching
- Example: Calculating total marks, arranging names alphabetically

4. STORAGE

- Saving information for future use
- Methods: Files, computer hard drive, cloud
- Example: Keeping student records

5. RETRIEVAL (Output)

- Getting stored information when needed
- Forms: Screen, printed paper, audio
- Example: Teacher getting report cards

6. COMMUNICATION

- Sharing information with others
- Methods: Reports, email, presentations
- Example: Sending report cards to parents

TYPES OF DATA PROCESSING:

1. MANUAL DATA PROCESSING

Definition: Processing by hand without computers

Tools: Pen, paper, calculator, files, registers

Examples:

- Teacher marks attendance in register
- Shopkeeper writes sales in notebook
- Library cards filled by hand

Advantages:

- No electricity needed
- No expensive equipment
- Simple

Disadvantages:

- Very slow
- Many errors possible

- Tiring
- Hard to store large amounts

2. ELECTRONIC DATA PROCESSING

Definition: Processing using computers

Tools: Computers, software, databases, internet

Examples:

- School management system
- Bank ATM
- Supermarket barcode scanner
- Online registration

Advantages:

- Very fast
- Accurate
- Can handle large amounts of data
- Easy to search
- Automatic

Disadvantages:

- Expensive
- Needs electricity
- Requires training
- Can have technical problems

COMPARISON:

Feature	Manual	Electronic
Speed	Slow	Very fast
Accuracy	Many errors	Few errors
Cost	Cheap	Expensive
Storage	Large space	Small space
Retrieval	Slow	Instant

REAL-LIFE APPLICATIONS:

- 1. Education:** Student records, report cards, timetables
- 2. Healthcare:** Patient records, appointments, test results
- 3. Banking:** Account balance, transfers, ATM
- 4. Business:** Sales, inventory, customer data
- 5. Government:** Census, ID cards, elections

EVALUATION:

1. What is data processing?
2. Name the six stages of data processing.
3. Differentiate between manual and electronic data processing.
4. Give three advantages of electronic processing.
5. State three real-life uses of data processing.

ASSIGNMENT:

1. Explain the difference between data and information.
2. Describe the six stages of data processing.
3. Compare manual and electronic data processing.

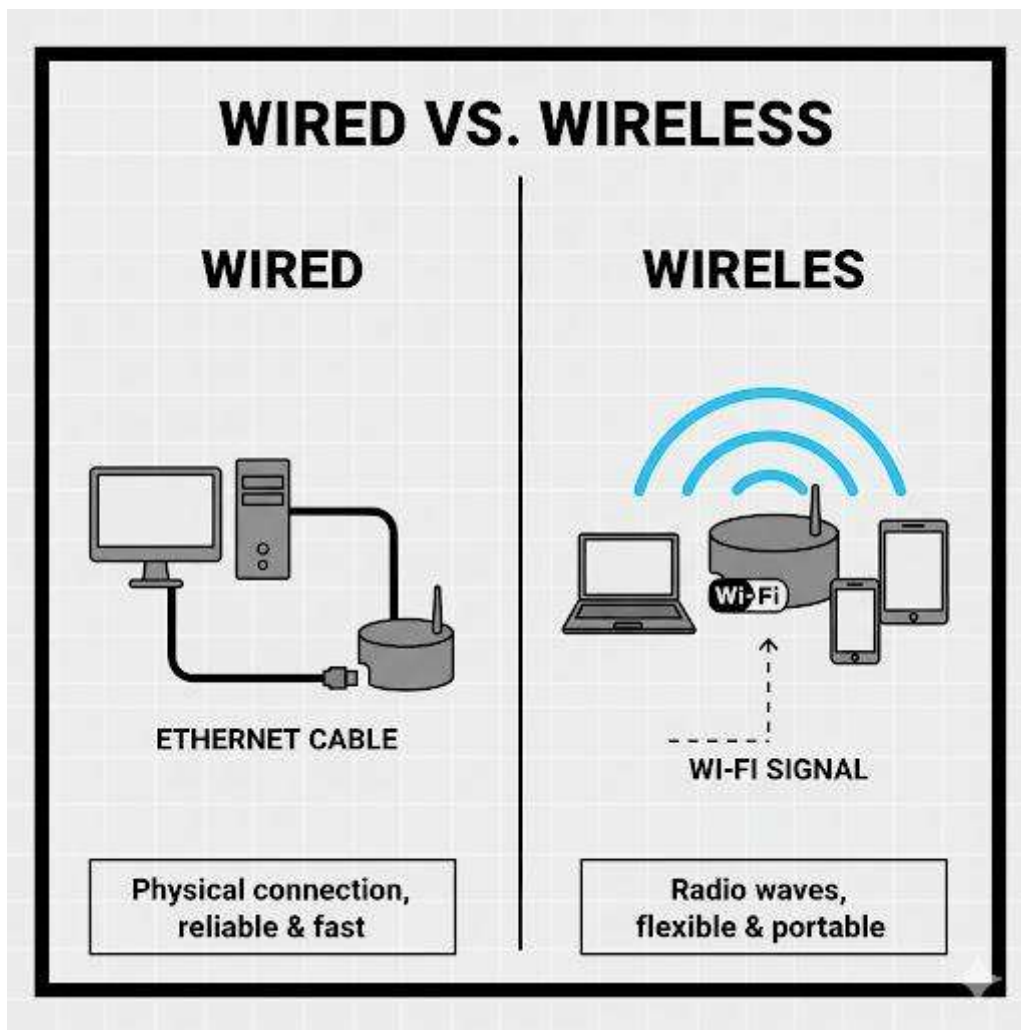
4. How is data processing used in schools and banks?



WEEK 4: BASIC COMPUTER COMMUNICATION

SUBTOPICS:

- MEANING AND IMPORTANCE OF COMMUNICATION
- TYPES: WIRED AND WIRELESS
- INTRODUCTION TO THE INTERNET



MEANING OF COMPUTER COMMUNICATION

Computer Communication: Sharing data between computers or devices.

Elements:

1. **Sender:** Device sending information
2. **Message:** Information being sent
3. **Medium:** Path for message (cable, Wi-Fi)
4. **Receiver:** Device receiving information
5. **Protocol:** Rules for communication

IMPORTANCE:

- Fast information sharing
- Global reach
- Cost-effective
- Collaboration
- Access to resources
- Entertainment and education

TYPES OF COMPUTER COMMUNICATION:

1. WIRED COMMUNICATION

Definition: Communication through physical cables.

Types of Cables:

A. Ethernet Cable:

- Most common for networks
- Connects computer to router
- Also called LAN cable

B. Fiber Optic Cable:

- Uses light to transmit data
- Very fast
- Used for high-speed internet

Advantages of Wired:

- Fast and reliable
- Secure
- Stable connection

Disadvantages:

- Limited mobility
- Installation needed
- Cables can be messy

2. WIRELESS COMMUNICATION

Definition: Communication without cables, using electromagnetic waves.

Types:

A. Wi-Fi

- Wireless internet network
- Router broadcasts signal
- Devices connect by entering password
- Range: 30-50 meters

Uses: Home internet, schools, cafés

Advantages:

- Mobility (move while connected)
- Multiple devices

- No cables

Disadvantages:

- Slower than wired
- Limited range
- Can be hacked if not secured

B. Bluetooth

- Short-range wireless (about 10 meters)
- Connects devices directly

Uses: Wireless headphones, mouse, keyboard, file transfer

C. Mobile/Cellular Networks (3G, 4G, 5G)

- Wireless through phone towers
- Mobile internet and calls

Advantages: Wide coverage, works while moving **Disadvantages:** Data charges, speed varies

D. Satellite

- Uses satellites in space
- Global coverage
- Used for GPS, satellite TV

COMPARISON:

Feature	Wired	Wireless
Speed	Faster	Slower
Mobility	Limited	High
Security	More secure	Less secure
Installation	Complex	Easy

INTRODUCTION TO THE INTERNET

What is the Internet? A global network of connected computers that share information.

How to Access Internet:

1. Device (computer, phone, tablet)
2. Internet connection (Wi-Fi, mobile data)
3. ISP account (MTN, Airtel, Glo, 9mobile)
4. Web browser (Chrome, Firefox, Edge)

What You Can Do:

1. **Communication:** Email, WhatsApp, video calls
2. **Information:** Google search, news, Wikipedia
3. **Education:** Online courses, YouTube tutorials
4. **Entertainment:** Netflix, YouTube, music, games
5. **Shopping:** Jumia, Konga, online stores
6. **Banking:** Check balance, transfer money

Important Terms:

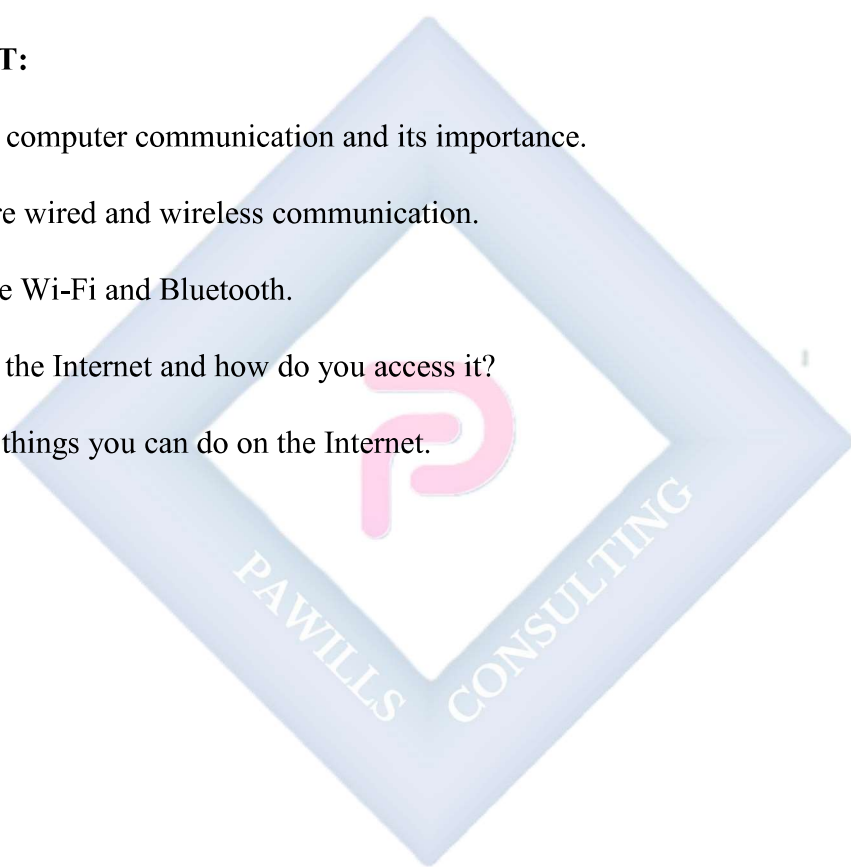
- **Website:** Collection of web pages (www.google.com)
- **URL:** Web address
- **Browser:** Software to view websites (Chrome)
- **Search Engine:** Helps find information (Google, Bing)
- **Download:** Copy from internet to device
- **Upload:** Send from device to internet
- **Streaming:** Watch/listen without downloading

EVALUATION:

1. What is computer communication?
2. Name three types of wireless communication.
3. Give three advantages of wired communication.
4. What is the Internet?
5. State five uses of the Internet.

ASSIGNMENT:

1. Explain computer communication and its importance.
2. Compare wired and wireless communication.
3. Describe Wi-Fi and Bluetooth.
4. What is the Internet and how do you access it?
5. List six things you can do on the Internet.



WEEK 6: INTRODUCTION TO COMPUTER NETWORKS

SUBTOPICS:

- DEFINITION OF A NETWORK
- COMPONENTS OF A NETWORK
- BENEFITS OF NETWORKING

WHAT IS A COMPUTER NETWORK?

Network: Two or more computers connected together to share resources and information.

Simple Example: Computers in your school computer lab connected together.

WHY CREATE NETWORKS?

- Share files and printers
- Share internet connection
- Communicate easily
- Work together on projects

TYPES OF NETWORKS:

1. LAN (Local Area Network)

- Small area (one building)
- School, office, home
- Fast and cheap

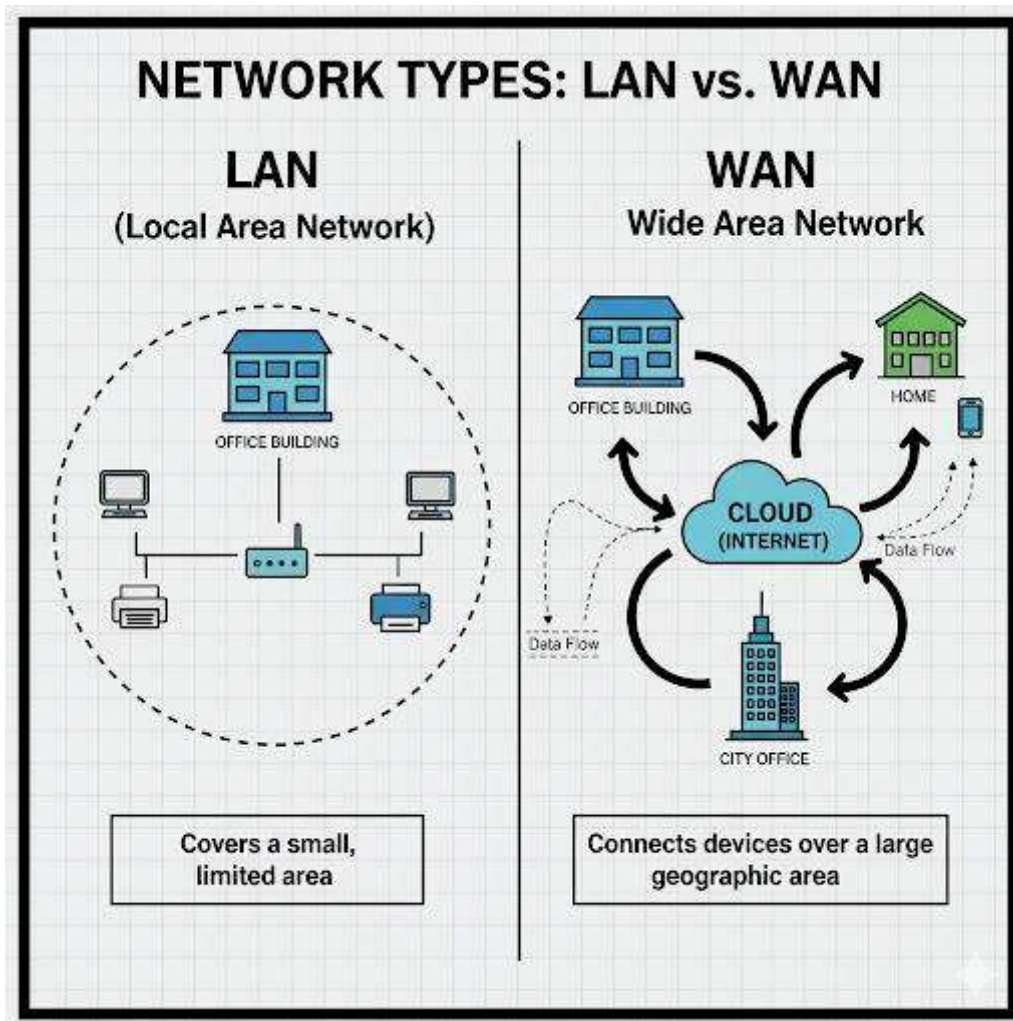
2. WAN (Wide Area Network)

- Large area (city, country, world)

- Internet is the largest WAN
- More expensive

3. MAN (Metropolitan Area Network)

- Covers a city
- Between LAN and WAN



COMPONENTS OF A NETWORK:

1. COMPUTERS/DEVICES

- Desktops, laptops, tablets, phones
- Called "nodes" or "hosts"

2. NETWORK INTERFACE CARD (NIC)

- Hardware inside computer
- Allows computer to connect to network
- Has unique MAC address

3. CABLES

- **Ethernet Cable:** Most common
- Connects devices together

4. ROUTER

- Connects network to internet
- Directs data to correct destination
- Wi-Fi routers broadcast wireless signal

5. SWITCH

- Connects multiple devices
- Allows them to communicate

6. MODEM

- Converts signals for internet connection
- Often combined with router

7. SERVER

- Powerful computer that provides services
- Stores files, manages network

8. NETWORK SOFTWARE

- Operating system
- Network protocols (rules)

- Security software

HOW A NETWORK WORKS (Simple):

1. Computer A wants to send file to Computer B
2. File broken into small packets
3. Packets travel through network (cables/Wi-Fi)
4. Switch/router directs packets to Computer B
5. Computer B receives and reassembles file
6. All happens in seconds!

BENEFITS OF NETWORKING:

1. RESOURCE SHARING

- **Printers:** One printer for many computers
- **Files:** Share documents easily
- **Internet:** One connection for all
- **Software:** Share programs

Example: School computer lab - 20 computers share 2 printers

2. COMMUNICATION

- Send messages instantly
- Email within organization
- Video conferencing
- Group chats

3. DATA STORAGE

- Central storage on server
- Access files from any computer
- Easy backup

4. COST SAVINGS

- Share expensive devices (printers, scanners)
- One internet connection
- Shared software licenses

5. COLLABORATION

- Work together on projects
- Share ideas
- Real-time updates

6. SECURITY

- Central backup
- Better virus protection
- Controlled access
- Monitor network activity

7. FLEXIBILITY

- Access from anywhere in network
- Add new devices easily

NETWORK TOPOLOGIES (Arrangement):

1. BUS TOPOLOGY

- All computers on single cable

- Simple and cheap
- If cable breaks, network fails

2. STAR TOPOLOGY

- All computers connect to central hub/switch
- Most common
- If one cable breaks, others still work

3. RING TOPOLOGY

- Computers in circular arrangement
- Data travels in one direction

NETWORK SECURITY:

- Passwords and user accounts
- Firewall (blocks unauthorized access)
- Antivirus software
- Encryption (scrambles data)

EVALUATION:

1. What is a computer network?
2. Differentiate between LAN and WAN.
3. Name five components of a network.
4. State four benefits of networking.
5. What does a router do?

ASSIGNMENT:

1. Define computer network with examples.

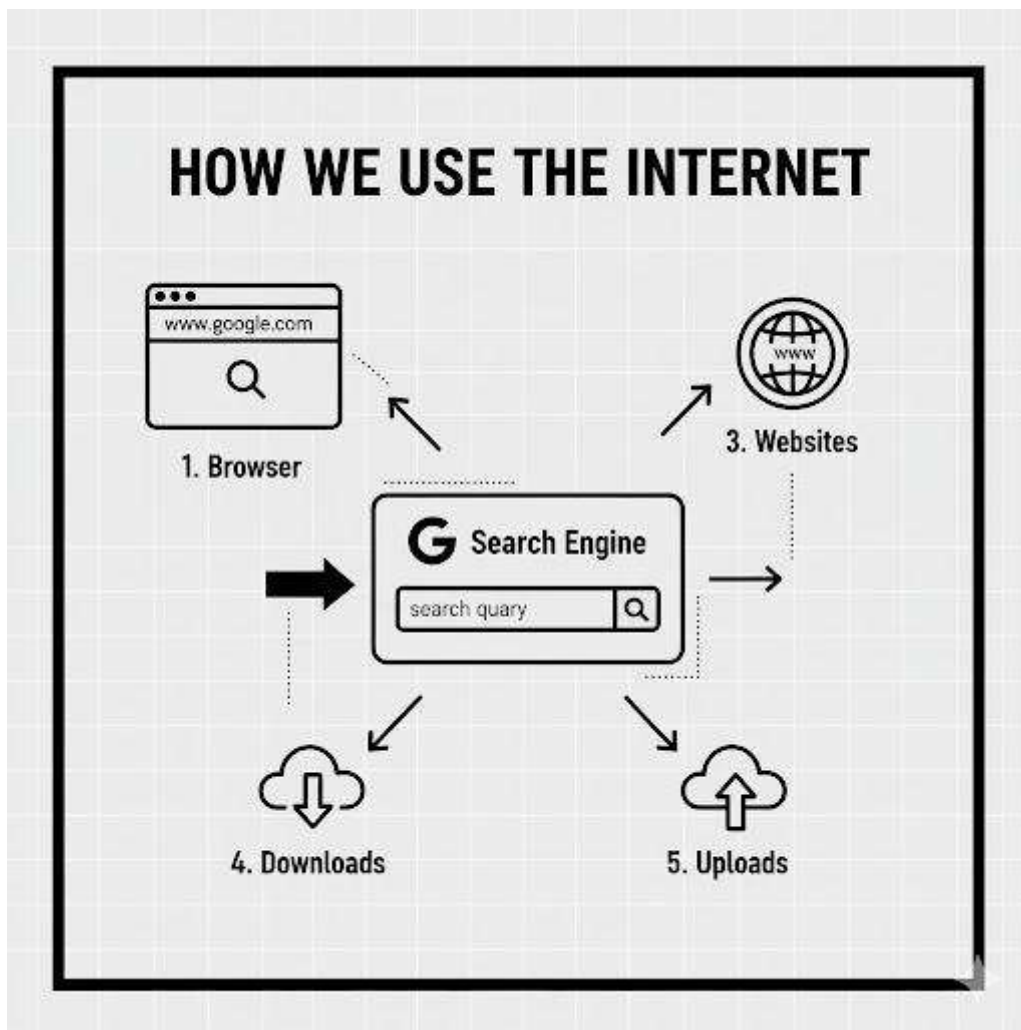
2. Describe six components of a network.
3. Explain five benefits of networking.
4. What is the difference between LAN, WAN, and MAN?
5. Draw and explain star topology.



WEEK 8: BASIC INTERNET CONCEPTS

SUBTOPICS:

- MEANING OF INTERNET
- COMMON TERMS
- RESPONSIBLE INTERNET USE



MEANING OF INTERNET

Internet: Global network of connected computers sharing information.

Also called: The Net, The Web, Cyberspace

Brief History:

- Started in 1960s (ARPANET)
- Public in 1990s
- World Wide Web created by Tim Berners-Lee (1989)
- Now billions of users worldwide

COMMON INTERNET TERMS:

1. WEBSITE

- Collection of web pages
- Has unique address
- Example: www.bbc.com

2. URL (Uniform Resource Locator)

- Web address
- Example: <https://www.google.com>
- **Parts:**
 - **https://:** Protocol (secure)
 - **www:** World Wide Web
 - **google:** Domain name
 - **.com:** Extension (commercial)

3. BROWSER

- Software to view websites
- **Examples:** Chrome, Firefox, Edge, Safari

How to use:

1. Open browser
2. Type URL in address bar
3. Press Enter
4. Website loads

4. SEARCH ENGINE

- Helps find information online
- **Examples:** Google, Bing, Yahoo

How to search:

1. Go to google.com
2. Type what you want to find
3. Press Enter or click Search
4. Results appear

5. HOMEPAGE

- First page of a website
- Main page

6. HYPERLINK (Link)

- Clickable text/image
- Takes you to another page
- Usually blue and underlined

7. DOWNLOAD

- Copy file from internet to your device
- Example: Download a song, document

8. UPLOAD

- Send file from your device to internet
- Example: Upload photo to Facebook

9. BANDWIDTH

- Speed of internet connection
- Measured in Mbps (Megabits per second)
- Higher = faster

10. STREAMING

- Watch/listen without downloading
- Example: YouTube videos, Spotify

11. ONLINE/OFFLINE

- **Online:** Connected to internet
- **Offline:** Not connected

12. ISP (Internet Service Provider)

- Company providing internet
- Nigeria: MTN, Airtel, Glo, 9mobile

13. EMAIL (Electronic Mail)

- Sending messages online
- Example: Gmail, Yahoo Mail

14. SOCIAL MEDIA

- Websites for connecting with people
- Example: Facebook, Instagram, Twitter, TikTok

15. CLOUD

- Storage on internet servers

- Access from anywhere
- Example: Google Drive, iCloud

TYPES OF WEBSITES:

- 1. Educational:** Schools, online courses (Khan Academy)
- 2. News:** BBC, CNN, Punch, Vanguard
- 3. Entertainment:** YouTube, Netflix, Spotify
- 4. Social Media:** Facebook, Instagram
- 5. E-commerce:** Jumia, Konga, Amazon
- 6. Search Engines:** Google, Bing
- 7. Government:** Official government sites (.gov.ng)

USING A BROWSER:

Basic Browser Functions:

Address Bar:

- Type web address
- Shows current URL

Back/Forward Buttons:

- Go to previous/next page

Refresh Button:

- Reload current page

Bookmarks/Favorites:

- Save websites you visit often

Tabs:

- Open multiple websites at once

Search Bar:

- Quick search without going to search engine

History:

- See websites you visited before

RESPONSIBLE INTERNET USE:**1. USE FOR GOOD PURPOSES**

- Educational research
- Learning new skills
- Positive communication
- Creative projects

2. DON'T:

- Cyberbully others
- Cheat on homework
- Visit inappropriate sites
- Spread false information
- Waste too much time

3. TIME MANAGEMENT

- Set limits
- Balance with other activities
- Don't use late at night

4. RESPECT OTHERS

- Be polite online
- No mean comments
- Respect privacy
- Don't share others' photos without permission

5. VERIFY INFORMATION

- Not everything online is true
- Check multiple sources
- Be critical

6. ASK PERMISSION

- Ask parents before:
 - Creating accounts
 - Downloading
 - Making purchases
 - Sharing personal information

INTERNET ETIQUETTE (Netiquette):

1. Use proper language (no insults)
2. Don't type in ALL CAPS (shouting)
3. Think before posting
4. Give credit to sources
5. Respect copyright

EVALUATION:

1. What is the Internet?
2. Explain: Website, URL, Browser, Search Engine
3. What is the difference between download and upload?
4. Name three types of websites with examples.
5. State four rules for responsible internet use.

ASSIGNMENT:

1. Define Internet and explain how it works simply.
2. Explain five common internet terms.
3. Describe how to use a search engine.
4. What is responsible internet use? Give five examples.
5. List six browser functions.

